

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 8: L12-L13

Question 1 (Lecture 12)

“If a car won’t start and battery is old, then replacing the battery might fix the issue”. What type of knowledge does this statement belong to?

- (a) Declarative
- (b) Procedural
- (c) Structural
- (d) Heuristic
- (e) Meta

Question 2 (Lecture 12)

Which of the following statements represent meta knowledge?

- (a) “The Eiffel Tower is in Paris.”
- (b) “The steps to solve a quadratic equation are: 1) rearrange terms, 2) apply quadratic formula.”
- (c) “If the reasoning process takes too long, switch from model checking to resolution.”
- (d) “The ground is wet. It possibly rained last night.”

Question 3 (Lecture 12)

In which scenario is probabilistic knowledge representation most useful?

- (a) When all facts are certain and must be represented with strict rules.
- (b) When handling noisy sensor data with uncertain outcomes.
- (c) When representing fixed hierarchical relationships, like a taxonomy of animals.
- (d) When encoding concepts as high-dimensional vectors for natural language processing.

Question 4 (Lecture 12)

What type of reasoning does propositional logic use?

- (a) Deductive
- (b) Inductive
- (c) Abductive

Question 5 (Lecture 13)

A single proposition symbol can represent a compound proposition.

- (a) True
- (b) False

Question 6 (Lecture 13)

Which of the following is/are a well-formed formula in propositional logic?

- (a) $\alpha = \gamma \Leftrightarrow \delta$
- (b) $\neg\neg\alpha$, provided that α is a well-formed formula
- (c) $\neg\alpha \vee \beta \Rightarrow \beta$, provided that α and β are both well-formed formulas.
- (d) $\alpha \wedge \sigma \wedge \neg\beta$
- (e) $\beta \models \alpha$, provided that α and β are both well-formed formulas.

Question 7 (Lecture 13)

Which of the following is/are correct?

- (a) $(A \wedge B) \models (A \vee B)$
- (b) $\alpha \models \beta$ if and only if $(\alpha \Rightarrow \beta)$ is a tautology
- (c) $\alpha \equiv \beta$ if and only if $(\alpha \Leftrightarrow \beta)$ is a tautology
- (d) $(A \wedge B) \wedge (A \Leftrightarrow B)$ is satisfiable

Question 8 (Lecture 13)

On an island, Knights always tell the truth and Knaves always lie.

Person A says: "Person B is a Knave."

Person B says: "We are both Knights."

- (i) Construct the knowledge base using propositional logic. Hint: you may use "sympy" library.
- (ii) Determine who is Knight and who is Knave.

Question 9 (Lecture 13)

Consider a simple 3x3 Wumpus World, where at least one Wumpus and at least one pit are located in some rooms. In the rooms directly (not diagonally) adjacent to the Wumpus, the agent will perceive a Stench. In the squares directly adjacent to a pit, the agent will perceive a Breeze.

- (i) Construct a simple knowledge base using propositional logic for this 3x3 Wumpus World. You may use "sympy" library. Use the following proposition symbol for

each $[x,y]$ location:

$P_{x,y}$: *There is a pit in $[x, y]$*

$W_{x,y}$: *There is a Wumpus in $[x, y]$*

$B_{x,y}$: *The agent perceives a breeze in $[x, y]$*

$S_{x,y}$: *The agent perceives a stench in $[x, y]$*

- (ii) Suppose the agent is now in location $[1,1]$, where neither Breeze nor Stench is perceived. Assume there is neither pit nor Wumpus in location $[1,1]$. Is there a Wumpus in location $[2, 2]$? Is there a pit in location $[2, 2]$?
- (iii) Continuing from part (ii), are adjacent cells of location $[1,1]$ safe? Suppose the agent moves to location $[2,1]$ and perceive a Breeze but no Stench. Are adjacent cells of location $[2,1]$ safe?